

1 Solution — GIAM 6.5

Problem 3

1.1 Problem

The natural logarithm function $\ln(x)$ is defined by the definite integral

$$\ln(x) = \int_{t=1}^x \frac{1}{t} dt, \quad x > 0.$$

Show that $\ln(x)$ is strictly increasing on its entire domain $(0, \infty)$.

1.2 Setup

A function f defined on a set D is **strictly increasing** if

$$a, b \in D, \quad a < b \implies f(a) < f(b).$$

So we must show: for every pair $0 < a < b$, $\ln(a) < \ln(b)$.

The integrand $1/t$ is continuous and strictly positive on all of $(0, \infty)$. That single fact will drive the entire proof.

1.3 The Argument

Geometrically, $\ln(b) - \ln(a)$ is the area under the curve $y = 1/t$ between $t = a$ and $t = b$. Since the integrand is strictly positive throughout $(0, \infty)$, that strip has positive area whenever $a < b$ — and the algebraic argument below just makes this rigorous.

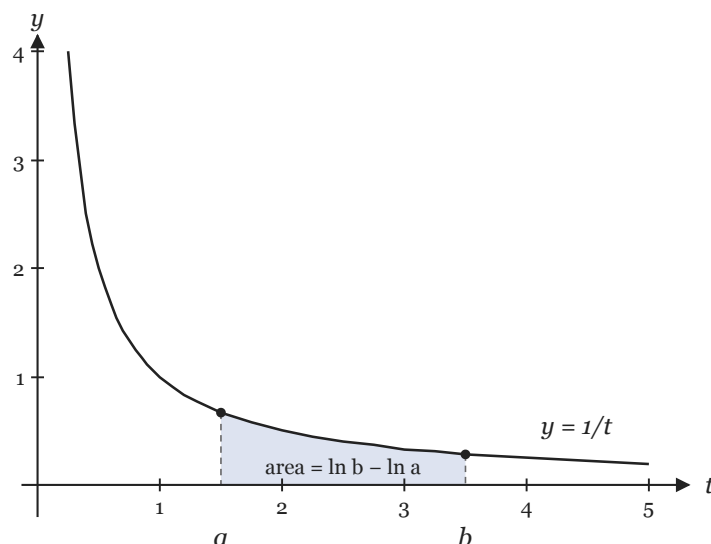


Figure 1.1: The integrand $y = 1/t$ is positive on $(0, \infty)$; the shaded strip from $t = a$ to $t = b$ has area $\ln(b) - \ln(a) > 0$.

Fix $0 < a < b$. By additivity of the integral over adjacent intervals,

$$\int_1^b \frac{1}{t} dt = \int_1^a \frac{1}{t} dt + \int_a^b \frac{1}{t} dt,$$

so the difference of the two logarithms is itself a definite integral:

$$\ln(b) - \ln(a) = \int_1^b \frac{1}{t} dt - \int_1^a \frac{1}{t} dt = \int_a^b \frac{1}{t} dt.$$

(The identity holds regardless of whether a, b lie above or below 1; the convention $\int_p^q = -\int_q^p$ keeps the bookkeeping consistent.)

Now bound the integrand from below. For every $t \in [a, b]$,

$$0 < a \leq t \leq b,$$

and reciprocating (which reverses inequalities for positive numbers) gives

$$\frac{1}{t} \geq \frac{1}{b} > 0.$$

So the integrand exceeds the positive constant $1/b$ throughout $[a, b]$, and the integral is correspondingly bounded:

$$\int_a^b \frac{1}{t} dt \geq \int_a^b \frac{1}{b} dt = \frac{b-a}{b} > 0.$$

Therefore

$$\ln(b) - \ln(a) \geq \frac{b-a}{b} > 0,$$

i.e., $\ln(a) < \ln(b)$. Since this holds for every $0 < a < b$, $\ln$ is strictly increasing on $(0, \infty)$. ■

1.4 Remark — The Calculus View

The Fundamental Theorem of Calculus gives an alternative, even shorter proof. Since $1/t$ is continuous on $(0, \infty)$, the FTC says

$$\frac{d}{dx} \ln(x) = \frac{1}{x}.$$

For every $x > 0$, $1/x > 0$, so the derivative of $\ln$ is strictly positive on its entire domain. Any function with strictly positive derivative on an interval is strictly increasing there — so $\ln$ is strictly increasing on $(0, \infty)$.

Both arguments rest on the same underlying fact: $1/t > 0$ for every $t > 0$. The integral-difference proof above extracts this directly from the definition of $\ln$; the FTC proof routes it through differentiation. Either way, the strict positivity of the integrand on $(0, \infty)$ is the engine behind the strict monotonicity of $\ln$.